

AI4SE and SE4AI: A Research Roadmap

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■ ABSTRACT

In 2019, the Research Council of the Systems Engineering Research Center (SERC), a US Defense Department sponsored University Affiliated Research Center (UARC), developed a roadmap structuring and guiding artificial intelligence (AI) and autonomy research. This paper presents that roadmap and key underlying Digital Engineering transformation aspects both enabling traditional systems engineering practice automation (AI4SE), and encourage new systems engineering practices supporting a new wave of automated, adaptive, and learning systems (SE4AI).

■ **KEYWORDS:** systems engineering, artificial intelligence, machine learning, automation, research

INTRODUCTION

Systems engineering is undergoing a digital transformation that will lead to transformational advances facilitating systems engineering use of artificial intelligence (AI) and machine learning (ML) technology to automate many routine engineering tasks. At the same time applying AI, ML, and autonomy to complex and critical systems encourages new systems engineering methods, processes, and tools. A 2019 future of systems engineering (FuSE) workshop, hosted by the International Council on Systems Engineering (INCOSE), first used the terms AI for systems engineering and systems engineering for AI to describe this dual transformation (Miller 2019). The “AI4SE” and “SE4AI” labels have become metaphors for an upcoming rapid evolutionary phase in the systems engineering Community. AI4SE applies augmented intelligence and machine learning techniques to support systems engineering practices. Goals in such applications include achieving scale in model construction and confidence in design space exploration. SE4AI applies systems engineering methods to learning-based systems’ design and operation. Key research application areas include developing principles for learning-based systems design, life cycle evolution models, and model curation methods.

To better understand and focus on this evolution, the Systems Engineering Research Center’s (SERC) research council,

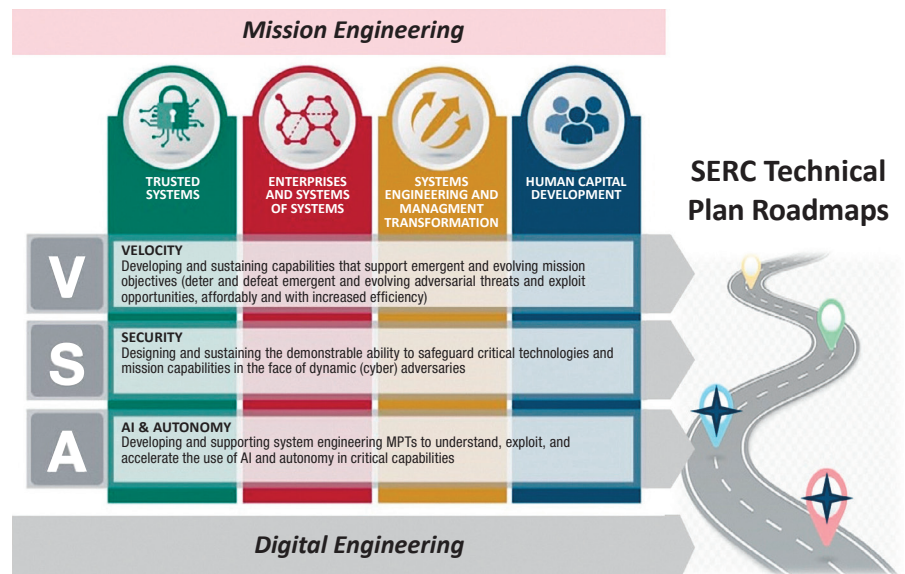


Figure 1. SERC research areas and missions

a US Defense Department sponsored university affiliated research center (UARC), developed a roadmap to structure and guide research in Artificial Intelligence (AI) and autonomy. This paper presents that roadmap. The SERC research strategy aligns three mission areas supported by four research areas, shown in Figure 1.

The research areas are enterprises and systems of systems (ESOS), trusted systems (TS), systems engineering and systems management transformation (SEMT), and human capital development (HCD). The

mission areas the SERC is addressing are:

- **Velocity:** Developing and sustaining timely capabilities supporting emergent and evolving mission objectives (deter and defeat emergent and evolving adversarial threats and exploit opportunities affordably and with increased efficiency).
- **Security:** Designing and sustaining the demonstrable ability to safeguard critical technologies and mission capabilities in the face of dynamic (cyber) adversaries.

- Artificial Intelligence (AI) and Autonomy: Developing and supporting system engineering methods, processes, and tools to understand, exploit, and accelerate AI and autonomy use in critical capabilities.

Digital engineering, which transforms systems engineering from document-based methods and artifacts to linked digital data and models, enables these.

The SERC technical plan, which outlines a 5-year vision for each research area, guides the mission and research areas (Systems Engineering Research Center 2019). Four roadmaps, developed collaboratively by the SERC research council over a 5-month effort in 2019, provide crosscutting mission area details (Systems Engineering Research Center 2019). Each roadmap has a set of verticals leading to a visionary outcome or set of outcomes, and a set of capabilities needed to meet those long-term outcomes. The listed capabilities reflect SERC research and other research, both active and prioritized by our sponsors and the systems engineering

community in general. By sharing this work, we hope to guide both SERC research and systems engineering transformations.

DIGITAL ENGINEERING AS THE ENABLER

Digital engineering forms the basis for all three SERC crosscutting missions and resulting research roadmaps. Systems engineering is in a transformation process based on the data use (an authoritative source of truth) and collaboration using models (collaborative integrated modeling environments). The SERC developed a digital engineering research roadmap which aligns with the five goals of our DoD sponsor's strategy: 1) model use for decision making; 2) the authoritative source of truth (AST); 3) technological innovation; 4) collaborative environments; and 5) workforce and cultural evolution (Zimmerman 2017). Figure 2 shows this roadmap.

Digital engineering's progression begins with data integration in the AST, then semantic model integration. We expect to see advances in Augmented Intelligence – model use and “big data” bringing auto-

mation to engineering processes, system quality, and certification. Augmented Intelligence forms the fourth “wave” of the digital engineering transformation in our roadmap and is AI4SE's core. There are multiple ongoing developmental efforts in this roadmap (yellow circles) and a few solid transition areas (blue circles). Summarized below are those relevant to AI4SE:

- Tool and Domain Taxonomies and Ontologies: we look to underlying engineering and programmatic data interoperability through future ontologies. Graph databases for linked data are becoming more prominent in model-based systems engineering (MBSE). Taxonomies provide the starting point for building ontologies, ultimately enabling AI-based reasoning on the underlying data. This is the transformational infrastructure in AI4SE.
- Semantic Rules: based on knowledge representations such as ontologies, semantic rules provide the basis for reasoning (AI) about completeness and consistency.

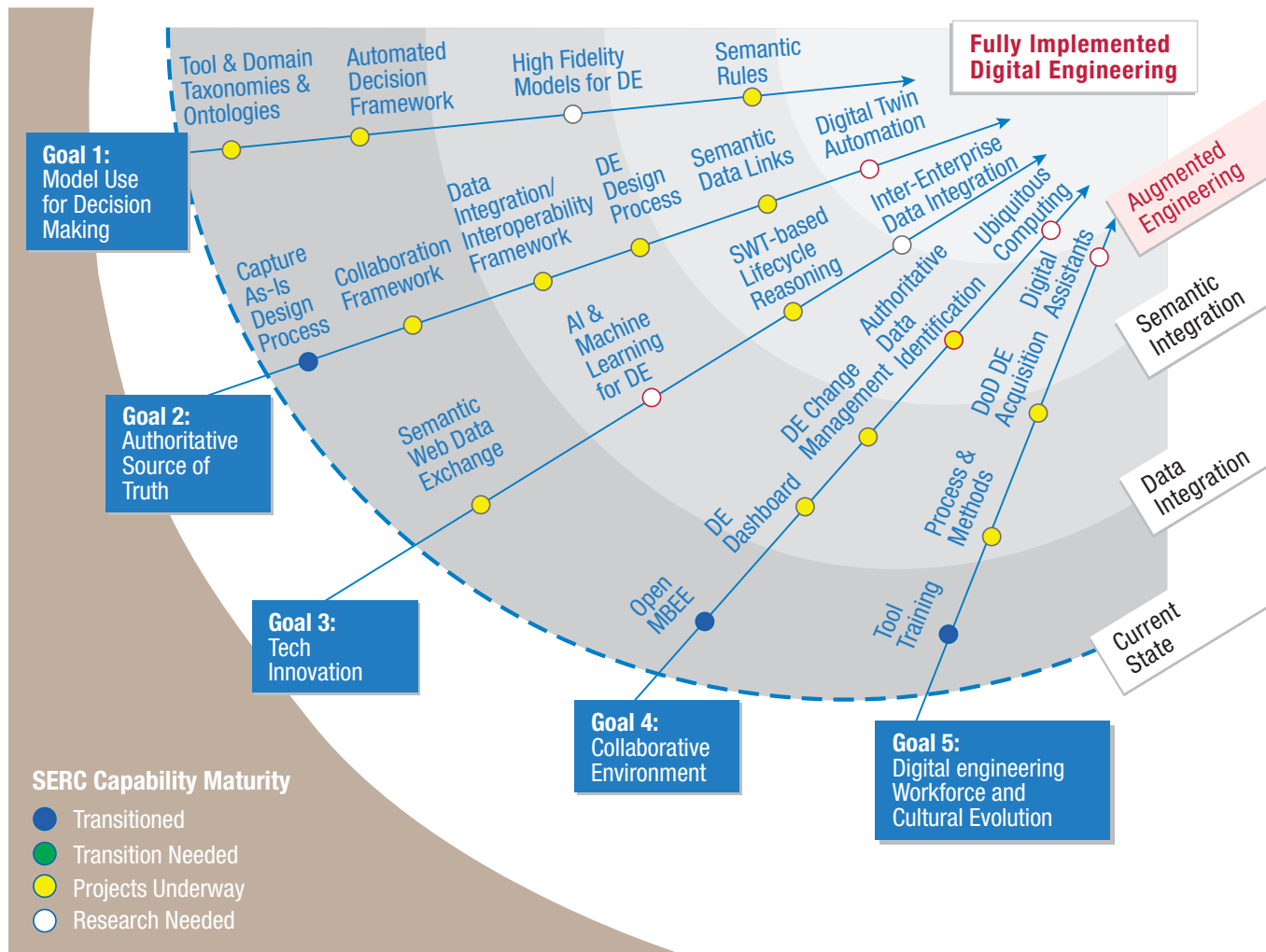


Figure 2. SERC digital engineering roadmap

In their article “Knowledge Representation with Ontologies and Semantic Web Technologies to Promote Augmented and Artificial Intelligence in Systems Engineering,” Blackburn et al. provide a detailed discussion of the importance of a semantic basis for engineering as an enabler for AI.

- **Inter-Enterprise Data Integration:** data/information seamlessly update/exchange continuously in “real-time” cutting across the entire enterprise (technical, manufacturing, cost, risk).
- **Automated Decision Frameworks:** combining ontologies, descriptive models, and analytics provide a decision making framework related to alternative analysis across any decision type, characterized by an objective hierarchy (basis for decision) and the underlying data.
- **Authoritative Data Identification:** automating how to find the “authoritative data,” assisted by AI/ML, and understanding what the user is looking for.
- **Digital Assistants:** trusted and automated AI guidance in engineering design and decision-making processes.
- **Digital Twin Automation:** this is the “end game” – fully dynamic virtual system copies built from the same models as the real systems running in parallel to physical systems and updating from the same data feeds as their real counterparts. This leads to opportunities for parallel learning in the real and simulated environments.

A FRAMEWORK FOR AI AND AUTONOMY RESEARCH

The SERC goal in artificial intelligence and autonomy is to lead the systems engineering transformation to dynamic processes leveraging the speed and rigor of rapidly evolving modeling, simulation, and analysis. Computational technologies and computational intelligence enable this transformation. The technical domain of AI, ML, and autonomy encompasses a broad range of emerging methods, processes, tools, and technologies. Although the technology will pass through “waves” led by digital data integration, semantic integration, and augmented engineering, we cannot yet define a concrete roadmap leading toward a central outcome. We can categorize research areas in an evolutionary framework expected to transform the engineering domain. Technological innovation’s “double S” curve (Figure 3) provides an effective categorization of systems engineering research contributions to emerging technology and their application. These include abstraction and high-level design methods, design for “X,” and design for test and certification, leading to ability

Technology Development → Digital Engineering → Mission Engineering

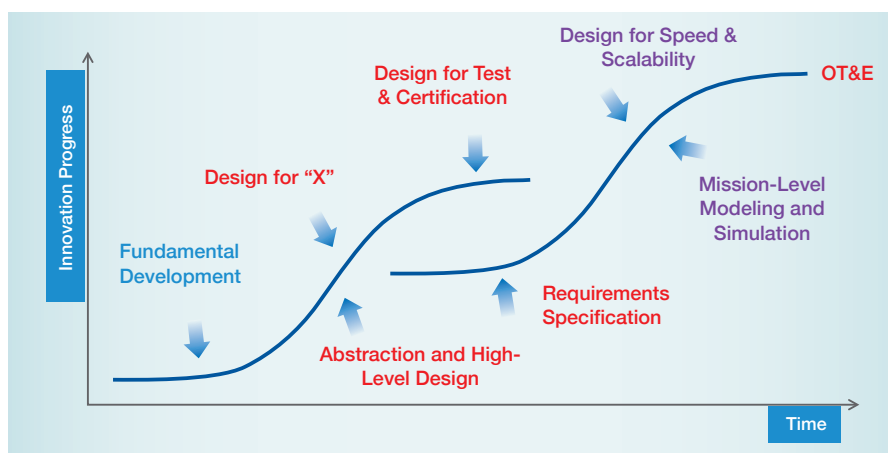


Figure 3. Systems engineering support to technological innovation

to specify the technology into requirements, tools to accelerate and scale design, modeling and simulation at the mission level, and finally to operational test and incorporation.

This domain’s advance, measured by the levels in our framework, form emergence “waves” as research transitions to practice. In the AI and autonomy roadmap, shown in Figure 4, green circles identify mature research areas where transition into the systems engineering domain is possible today, the yellow circles are ongoing systems engineering research activities, and the white circles are areas where systems engineering research should begin. The roadmap centers around four major verticals:

- **AI/ML Technology** – technological advances supporting use in systems;
- **Automation and Manned/Unmanned Teaming** – methods and tools ensuring beneficial and safe use of resulting automation; and
- **Systems Engineering Process Evolution** through Digital Engineering – the evolution of systems engineering process to learning technologies and automation and the transition to a digital engineering data driven basis for engineering which allows automation and learning.

Described below, organized by the four verticals, are the research areas:

AI and Machine Learning Technology

A continuing set of research needs in the basic technologies of AI/ML and autonomy both enabling AI4SE and evolving SE4AI processes are:

- **Multi-Modal AI:** holistic multi-modal data analysis (sensed, discussed, written, social, environmental) aiming to produce actionable intelligence for human decisions (systems engineering design decisions in this case). Most ML

applications today are narrow, focusing on single modalities. However, AI applications fusing multiple modalities have been in service for years. Linking ML-based learning to AI-based decision support algorithms provides one foundation of AI4SE.

- **Cognitive Bias:** a need to counter intentional or unintentional misled decision-making in AI systems, caused by the training data, adaptive learning over time, or intentional misinformation. Current narrow ML applications are subject to such biases. Augmented intelligence algorithms for SE must be trustworthy.
- **Hybrid Human/AI Systems:** automated machine reasoning agents helping humans understand complex data. Current AI/ML applications work well with human reasoning in narrow well-defined contexts. Highly automated complex systems (self-driving vehicles) face human versus machine control challenges. Both the AI4SE process, and new systems engineering processes defining interactions in SE4AI evolution require seamlessly integrating human and machine tasking and learning. In future systems engineering practice, human systems integration will no longer be a specialty area but will be front and center to system definition.

In his article “As Smart as a Human? Leveraging models of human intelligence to assess the intelligence of systems,” Brown provides a thoughtful approach to evaluate human and machine system “intelligence.”

- **Contextual Sensemaking:** research into AI perceiving and learning context, such as DARPA’s Third Wave initiatives (Launchbury 2015), will provide

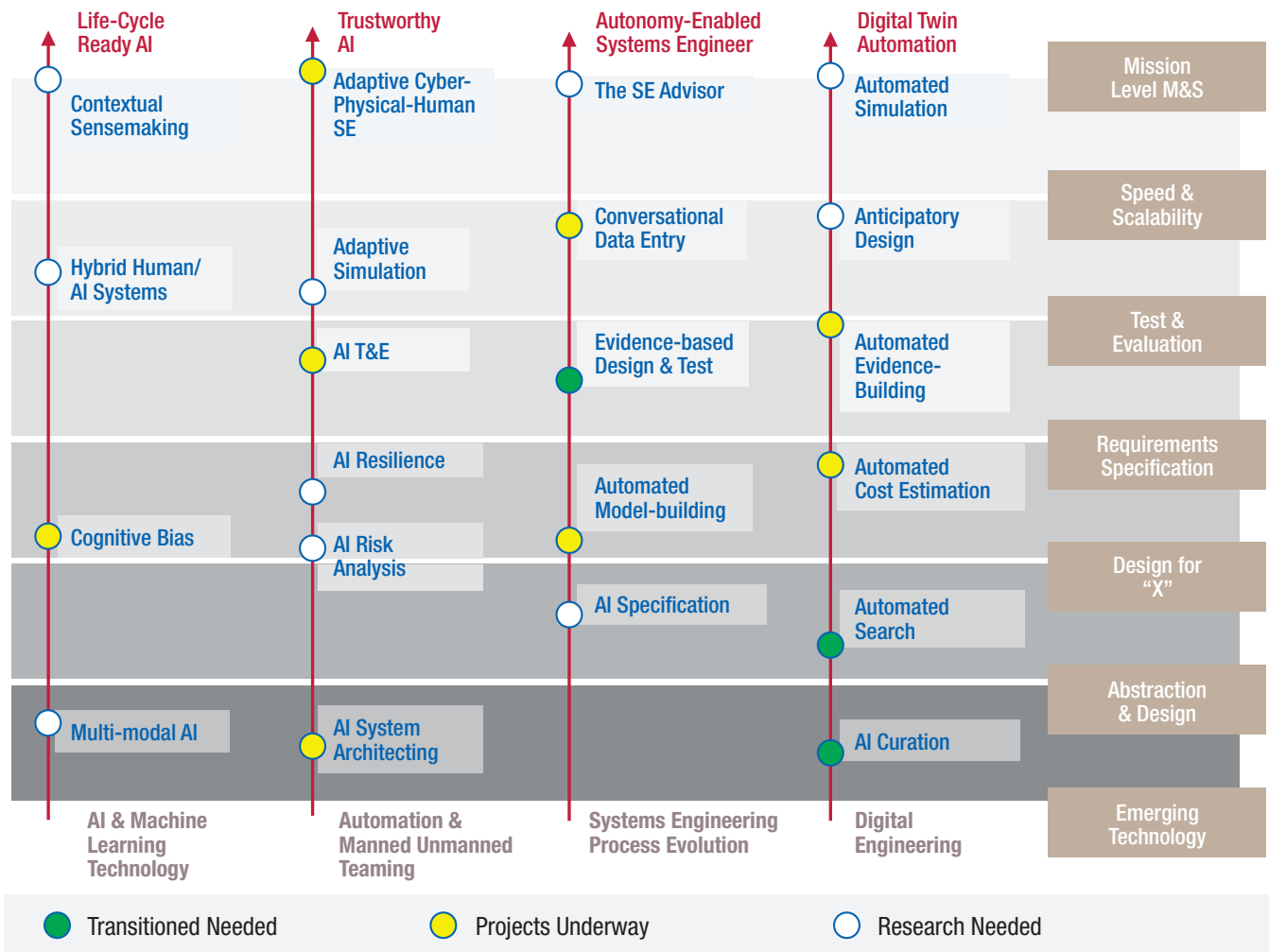


Figure 4. SERC AI and autonomy roadmap

foundational platforms for future hybrid human/AI systems.

- Life-Cycle Ready AI: the technology development vertical outcome is a body of new systems engineering methods and tools addressing increasingly complex learning and adapting systems with the rigor traditional to the systems engineering domain. We are beginning to see system level issues addressed in basic AI/ML applications research. Figure 5 details multiple research topics addressing dependable/safe AI produced by DeepMind, a leading company in basic AI research, (Ortega Maini and DeepMind 2018). AI safety researchers can categorize key topics in terms of traditional systems engineering and control areas, such as system modeling, stability, and resilience. However, SE4AI processes specific to the management and control AI/ML system life cycles remain lacking.

Automation and Manned-Unmanned Teaming

Most AI/ML application's goals are automating previously manual or partially automated human processes. Thus, automation and human interaction research (manned-unmanned

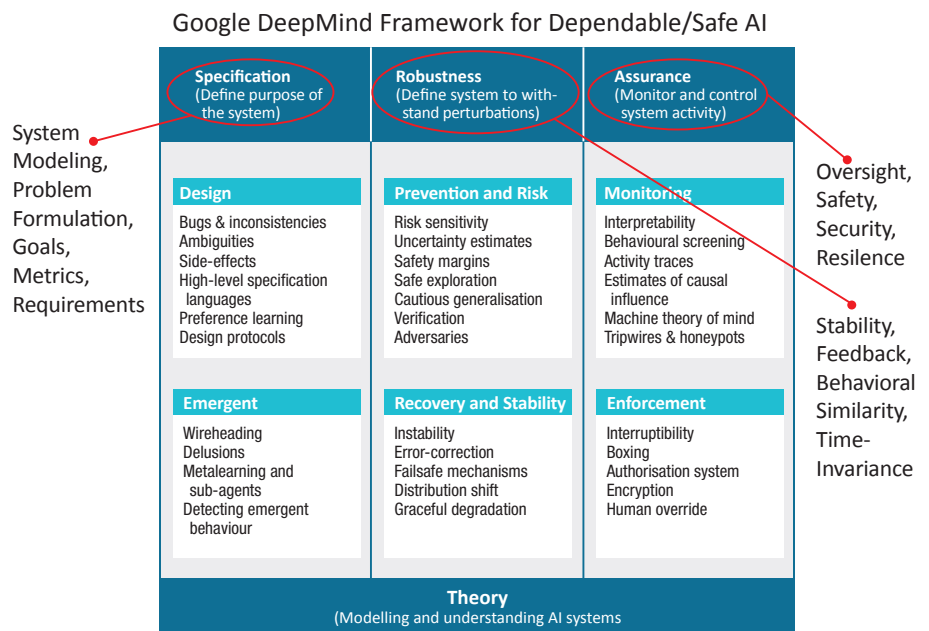


Figure 5. SE practices influencing current AI/ML research (adapted from (Ortega Maini and DeepMind 2018))

teaming) is an essential part of the systems engineering of these systems. Systems engineering included automation for many years, but systems adapting and learning in their deployment, particularly at scale, are a challenge for current systems engineering processes. Research will help develop the methods, processes, and tools that ensure these systems are trustworthy.

- **AI Risk Analysis:** methods, processes, and tools need to connect system risk analysis results with AI software modules related to those risks. A fundamental risk ontology for AI/ML systems needs development.
- **AI Resilience:** future systems should include resilience capabilities addressing AI related failure modes. This will involve system level behavior monitoring preventing the AI algorithms from performing operations outside pre-defined boundaries. research includes methods to evaluate AI/ML predictive algorithm generation (generally part of “explainable AI” research).
- **AI Test and Evaluation:** methods for addressing AI-related system test and evaluation addressing these systems’ ability to adapt and learn from changing deployment contexts.

In her article “Test and Evaluation for Artificial Intelligence,” Freeman reviews the challenges facing the test and evaluation community to evaluate learning systems.

- **Adaptive Simulation:** computer-based simulation and training supporting non-static objectives and/or goals (pick-up games, course of action analysis) necessary to provide contextual learning environments for these systems. Most simulations today derive from static decomposition and will evolve to include learning agents which evolve the simulation over time. Real and simulated co-learning (digital twins) will be a standard systems development form.
- **Adaptive Cyber-Physical-Human Systems:** future systems will employ different types of human and machine learning to flexibly respond to unexpected or novel situations, using plan and goal adjustment and adaptation, learning from experience, and continuous task adaptation. Research challenges in the area include simulations reflecting human behavior and learning, and methods to address different time domains in the real and machine worlds.

In his article “Exploiting Augmented Intelligence in Systems Engineering and Engineered Systems,” Madni discusses a

framework for incorporating Augmented Intelligence in both engineered systems and systems engineering processes.

- **Trustworthy AI:** resulting AI systems that self-adapt while maintaining rigorous safety and security and policy constraints are this research vertical’s primary outcome. Trust includes the machine’s ability to meet human behavior standards and the impact to human situational awareness as trust is turned over to the machine.
- **System Architecting of AI:** the system architecting process may change as automation grows into more complex systems. System architectures must support learning and adaptation, and more agile change processes. A primary component of future architecting will be parallel development and comparison of in vivo (real) and in silico (virtual) deployments.

In their article “Motivating a Systems Theory of AI,” Cody, Adams, and Beling discuss a system theoretic approach for architecting AI/ML systems.

Systems Engineering Process Evolution

Both AI/ML technology development and the needs to manage and control the resultant automation will drive Systems Engineering evolution. The following are some critical research areas driving the systems engineering disciplines.

- **AI Abstraction:** achieving AI4SE at scale will require library tools and methods abstracting AI algorithms and programming modules to engineering’s general use. Developing such abstractions transformed the microelectronics industry. Similar developments will bring AI/ML to a broader engineering domain.

In their article “A Systems Engineering Approach for Artificial Intelligence: Inspired by the VLSI Revolution of Mead & Conway,” Wade and Buenfil discuss a model using abstraction to bring AI/ML to the systems engineering community based on similar experience in the VLSI revolution.

- **AI specification:** a requirements specification and management process for adaptation and learning in systems does not exist today and needs developing. Coding environments will integrate much of this, building from today’s Devops environments to systems models. Static requirements specification will only partially support future systems.

- **Evidence-based Design and Test:** SE4AI requires formal methods and processes moving from explicit composition verification to evidence building strategies focusing on adaptation acceptance or prevention. The SE4AI domain needs assurance methods linked to validation methods. In the AI4SE practice, ML based tools to find and reason on evidence will be a key enabler.

In their article “Validation of AI-Enabled and Autonomous Learning Systems,” Collopy and Sitterle discuss the challenges of AI/ML system validation and suggest a framework for formal validation.

- **Conversational Data Entry:** human-computer interaction processes to convert natural language and other media to formal models will greatly improve knowledge transfer and consistency in future systems engineering tools.
- **The systems engineering Advisor:** we envision, as an outcome, a conversational system automating many mundane data entry, exploration, engineering calculation tasks, and many workflows.

In his article “AI as SE: Augmented Intelligence for Systems Engineers” Rouse reviews what this advisor might do and an architectural framework for its development.

- **Autonomy Enabled Systems Engineers:** in the long-term, system engineers become masters at deploying autonomy as a design variable and AI/ML as a design tool. Although not listed as a vertical in this roadmap, autonomy and AI skillset development in the engineering domain is quickly becoming a need.

Digital Engineering

Systems Engineering’s transformation to first a model-based discipline then to a data-driven discipline, as discussed in the digital engineering roadmap, will enable rapid evolution of AI/ML technologies into our tools. Some of the critical areas are:

- **AI Curation:** data management and digital engineering data and model curation is a core transformational need for systems engineering. Data curation will be necessary to support evolving AI capability application, and AI/ML tools will see extensive use in managing the data sets. Training data curation is a new role in systems engineering and will be critical to future systems.
- **Automated search, model-building, and cost estimation:** applying ML to historical data and relationships will greatly

improve systems engineering process speed and consistency. Applying statistical methods and AI algorithms in these activities has been present for a number of years, but ML technologies will greatly increase these tools' accessibility and use.

- Automated evidence building: certification process automation via models and quality assurance will be a helpful AI/ML evolution outcome, particularly for systems-of-systems and distributed development and test activities.
- Anticipatory design: an expected and desired outcome of increasing engineering practice automation is a tool anticipating system emergence (failures) from design & operational data. This should lead to more trustworthy systems.
- Automated Simulation: simulation use to train and evaluate ML in an emerging research area. This is a generative adversary network's evolution. System theoretical methods (system mathematical frameworks) for training AI/ML algorithms in complex systems is an emerging research area.
- Digital Twin Automation—Industry 4.0 initiatives envision real-time continuous learning from real system and shadow simulations. The digital twin evolution from generally static models to fully dynamic simulations, even at mission level, will transform not only the systems engineering process, but the products we develop.

VELOCITY

Velocity and agility are critical future system characteristics, both for the deployed system and the system developing and maintaining the deployed system. With the fusion of development in operations, DevOps, the delineation between these is disappearing. A research roadmap for velocity (Systems Engineering Research Center 2019) is the most difficult to articulate as current organizational practice and methodology implementations root it. One might ask where is the needed research? Velocity, with SERC's defense and other government sponsors, centers on three goals: architecting systems for continuous development and deployment, leading an agile transition across large government and contractor systems, and collaborative integrated modeling environment role as an enabler. These interweave with the AI/autonomy research thrusts (and their motivating needs) highlighted earlier.

Overall our vision is to enable systems engineering's transformation from sequential, document-driven, highly constrained practices toward much faster, flexible mission and enterprise-orient-

ed approaches enabled by advances in modeling, simulation, data-driven analysis, and artificial intelligence. This paper does not discuss the velocity roadmap in detail, but the primary research verticals in this area strive for application in two areas: improved mission engineering processes and adaptive system creation. Research areas include rapid development of systems as platforms, architecting these platforms for DevOps enabled systems and environments, and DevOps practice execution in the systems engineering process.

SECURITY

The SERC security roadmap (Systems Engineering Research Center 2019), also not presented in detail here, focuses on critical engineered systems such as cyber-physical systems, embedded systems, and weapon systems. These are critical, highly assured systems. The roadmap recognizes attributes such as security and resilience as critical system properties, and assurance as a process that yields an evidentiary case that a system is trustworthy with respect to the properties its stakeholders legitimately rely upon. Ongoing SERC security research focuses on three areas: (1) prevent, detect, and mitigate security vulnerabilities; (2) design, model, and conduct analysis of trustworthiness (safe and secure aspects) of complex cyber-physical system capabilities and behaviors; and (3) develop models, processes, and tools to assure the trustworthiness of system behaviors/ performance evolves increasingly driven by machine learning, autonomous capabilities, and manned-unmanned teaming.

Systems based on AI/ML are vulnerable to adversarial attacks on the learning algorithms. This might be through conventional cyber-attack vectors or through learning algorithm input data disruption (sensor data) via physical means in the environment. Additionally, the learning process' complexity and non-determinism mean,

even in the absence of adversarial action, field condition performance may diverge from testing performance.

Engineering for AI/ML resilience is beginning. Today, few understand how to design learning-based systems resilient to an intelligent adversary's attacks or to use, environment, and condition changes occurring over a critical system's long lifecycle. However, these goals are similar to engineering resilience against cyber threats where the aim is inhibiting adversarial action, enduring an attack by maintaining minimum mission capability, and finally restoring full functionality in a timely manner. As part of the security roadmap research, SERC has developed numerous design patterns and technical methods for detecting and responding to cyber-attacks, alongside methodologies and tools supporting cyber-resilient design and vulnerability assessment (Beling Horowitz and Fleming 2019)(Carter Adams Barkirtzis Sherburne Beling Horowitz and Fleming 2019). This engineering for cyber resilience will provide a foundation for the study of engineering for AI resilience.

SUMMARY

Roadmapping various research opportunities is important to communicate broad research goals and to connect researchers working on individual projects to larger impact opportunities. It also serves to help professional societies, government, industry, and academia focus their strategic goals. Systems engineering is transforming, enabled by digital engineering and shifted by the SE4AI and AI4SE concepts. These interlink with the need for methods, processes, and tools leading to adaptive, continually developed and deployed systems (velocity) while ensuring security and resilience are inherent in the outcomes. Perhaps the systems engineering community's biggest challenge is to adapt itself to these transformative concepts' challenges. ■

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